Neurobiology of Neuron

Nazia Shahab
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Nervous system comprises : neurons and neuroglia

Neurons --- situated in brain or center

Extensions of neurons --- **Axon,** are away from the center i.e.

Extended to the periphery

Receptor Terminal

- *Soma / Perikaryon*
- *Neurites: Dendrites & Axon*
- *Dendrites—Arborization*

Conductile Portion

- *Axon hillock*
- *Axon --- 1 or 2*
- *Myelin--- oligodendrocytes*
- *Nodes of Ranvier*

Synaptic Terminal

- *Arborization*
- *Synaptic ends---Terminal boutons*

Neuron

Structure & Function of nervous tissue

Have processes called as **neurites**

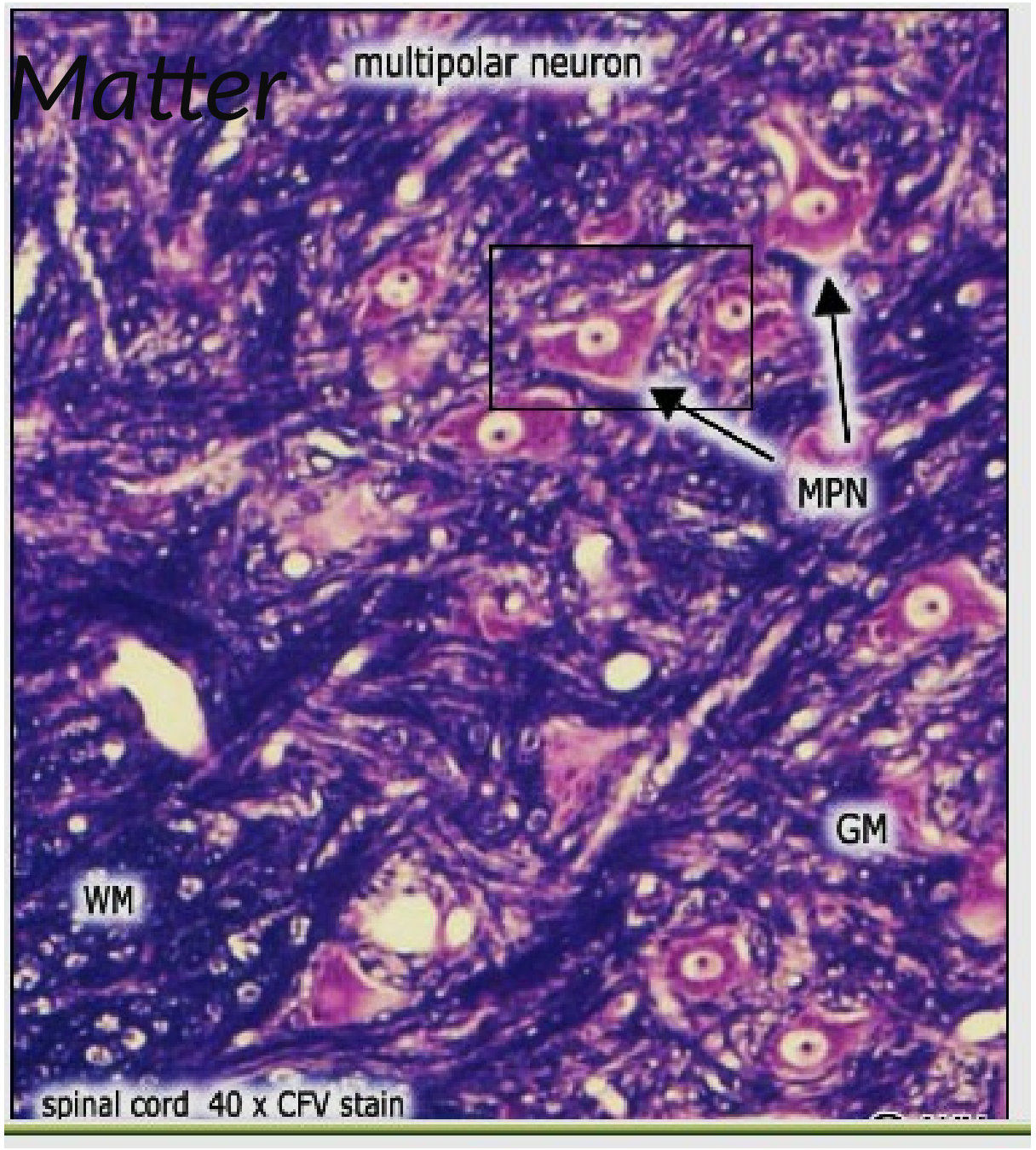
A neurite maybe: dendrite or axon

Like any cell i.e. cell organelles

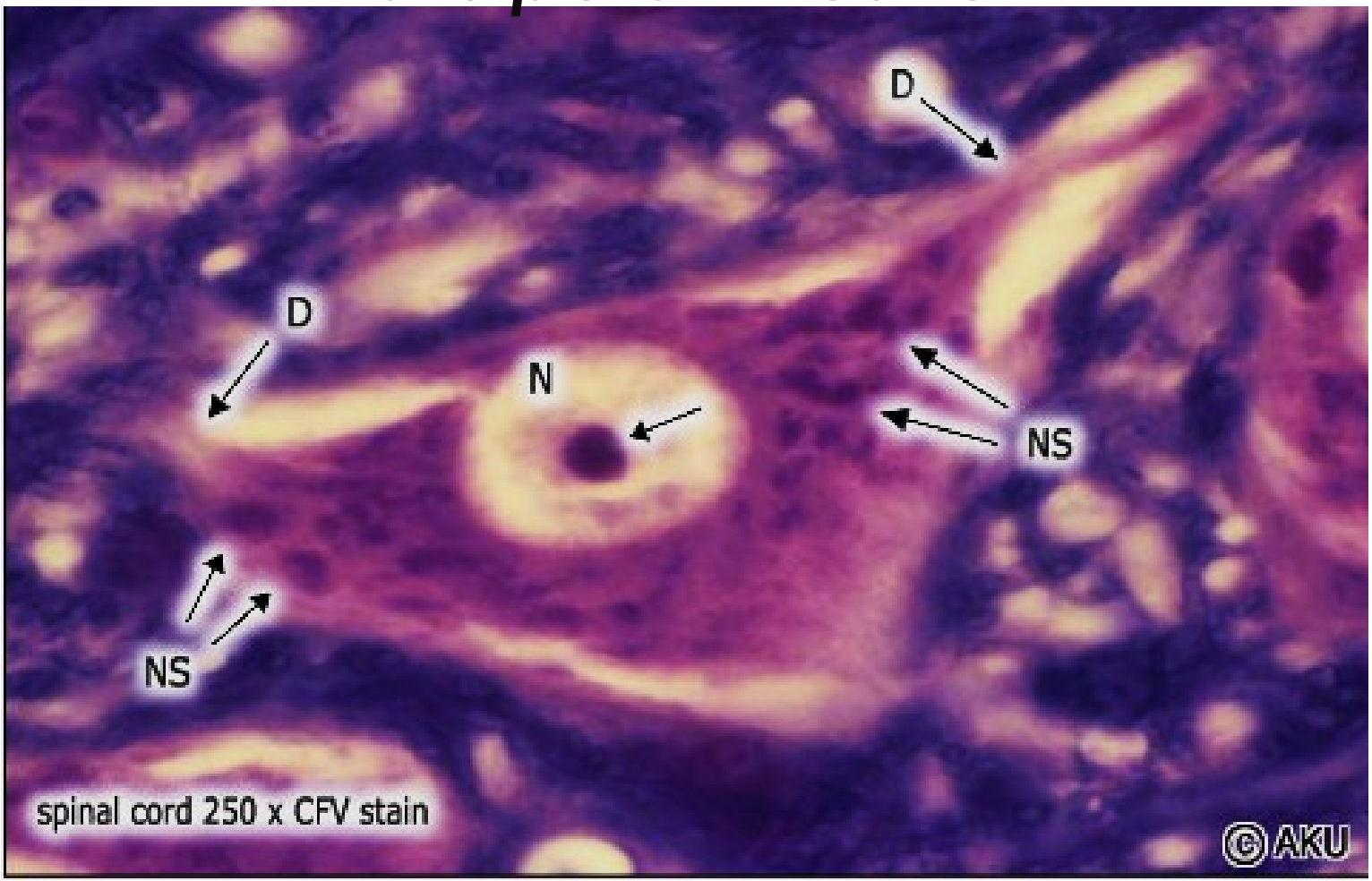
Neuron

- Membrane bound:
  - Nucleus
  - Nucleolus
  - RER
  - SER
  - Golgi Apparatus
  - Lysosomes
  - Mitochondria
- Non-membrane bound:
  - Neurofibril
  - Microtubule
  - Centrioles
  - Pigments e.g. Lipofuscin, melanin

Gray Matter



Multipolar neuron



Neurons

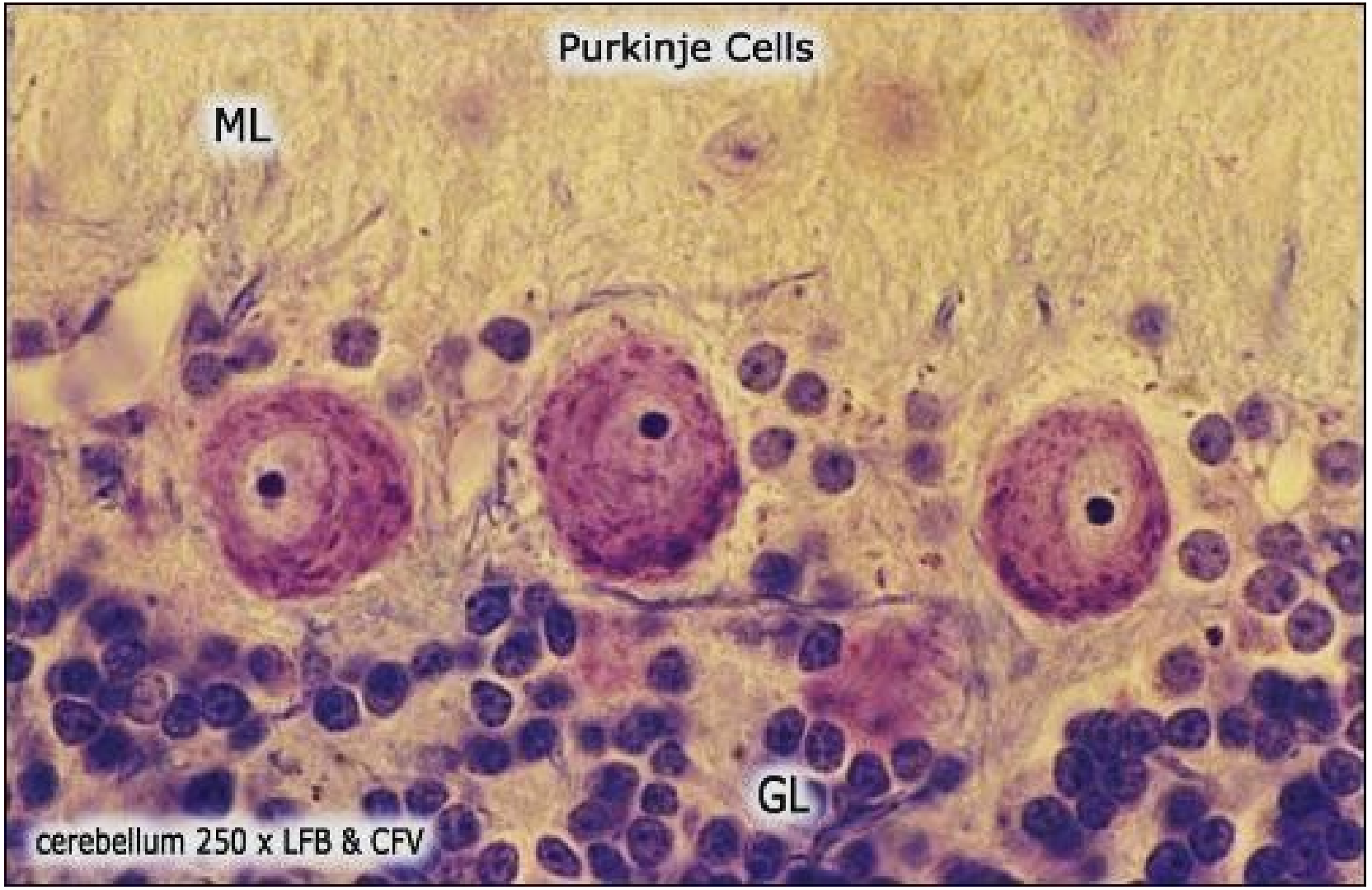
- According to no. of processes
  - Unipolar/Pseudo-unipolar
  - Bipolar
  - Multipolar

- According to shape
  - Stellate
  - Pyramidal
  - Fusiform
  - Horizontal

- According to length of processes
  - Golgi Tendon I
  - Golgi Tendon II

Types of Neurons – According to No. of Processes

Types of Neurons – According to Shape of Cells


Purkinje Cells
ML
GL
cerebellum 250 x LFB & CFV

Types of Neurons – According to Shape of Cells

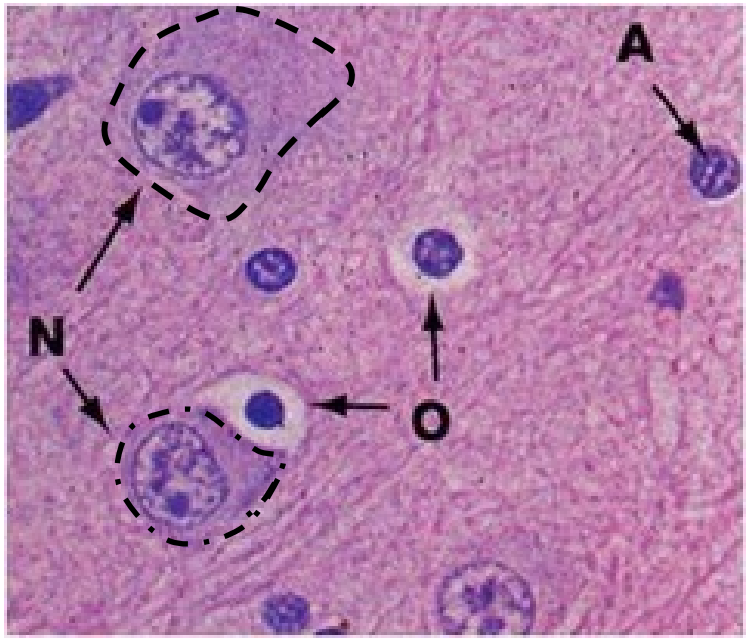
Neuron cell body --- gray appearance

Gray Matter --- Neuron cell body, glial processes

Or

Unmyelinated nerve

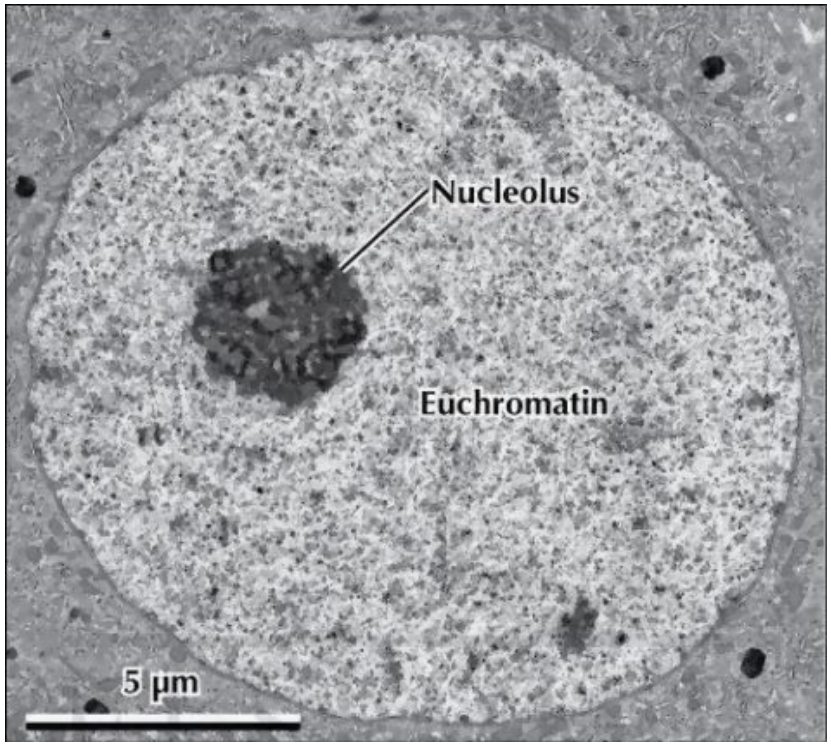
White matter --- neuronal processes with surrounding myelin sheath




cerebral cortex
cerebrum 25 x Golgi stain
© AKU

Neuronal Synapses

Nucleus



Quiz

- What are the components of nervous tissue?
- How many parts are there in a neuron?
- What are Nissl's substance/ granules?
- What is Golgi tendon I?
- What is Golgi Tendon II?
- What comprises gray matter?
- What comprises white matter?
- Name some of the synapses formed by a neuron

Thank you

- References:

  Snell's Clinical Anatomy 5th edition

  Netter's Essential Histology

  Barr's The Human Nervous System Anatomical Viewpoint, 9th Edition

- Courtesy:

  AKU's Digital Atlas of Microscopic Anatomy

  Wheater's Functional Histology 4th Edition

  www. Google.com